

Skills Summary

Separable Differential Equations with Initial Conditions

Use this reference while completing your worksheet or when studying.

Skill 1 — Separating and integrating

When is an equation separable? When the right-hand side factors as (something in x) times (something in y):

$$\frac{dy}{dx} = f(x)g(y) \implies \frac{dy}{g(y)} = f(x)dx.$$

Integrate each side in its own variable. **One** constant of integration is enough — write $+C$ on the x side. The two integrals you will meet most often are $\int \frac{dy}{y} = \ln|y|$ and $\int y dy = \frac{1}{2}y^2$.

Example: Find the general solution of $\frac{dy}{dx} = 6x^2y$.

Step 1. The right side is already a product of $6x^2$ and y , so the equation is separable: divide by y and multiply by dx to put each variable with its own differential.

Step 2. $\int \frac{dy}{y} = \int 6x^2 dx$, and the two antiderivatives are $\ln|y|$ and $2x^3$.
 $\implies \ln|y| = 2x^3 + C$

Try it: Find the general solution of $\frac{dy}{dx} = 8x^3y$.

check: $y = e^{2x^4}$

Skill 2 — Finding C from the start value

Order of operations that saves algebra: integrate first, *then* substitute the initial values, *then* solve for the constant, *then* rearrange into $y = f(x)$. Exponentiating $\ln|y| = F(x) + C$ gives $y = Ae^{F(x)}$ with $A = e^C$, and for a positive initial value A is simply y at the starting point.

Example: Solve $\frac{dy}{dx} = 4xy$ with $y(0) = 2$, then find $y(1)$.

Step 1. An initial value is given, so integrate the separated equation and use the starting point to pin the constant before evaluating anything.

Step 2. $\int \frac{dy}{y} = \int 4x dx$ gives $\ln|y| = 2x^2 + C$, so $y = Ae^{2x^2}$; the condition $y(0) = 2$ forces $A = 2$, hence $y = 2e^{2x^2}$.
 $\implies y(1) = 2e^2$, so **$y(1) \approx 14.778$**

Try it: Solve $\frac{dy}{dx} = 6xy$ with $y(0) = 3$, then find $y(1)$.

check: $y(1) \approx 25.78$

Skill 3 — Using the particular solution

Two questions, one solution. Once $y = f(x)$ is known: to find a *value*, substitute the given input; to find *when* a value occurs, set $f(x)$ equal to it and solve. For $y = Ae^{kx}$ the second question always ends in a logarithm: $Ae^{kx} = B$ gives $x = \frac{1}{k} \ln\left(\frac{B}{A}\right)$.

Example: $\frac{dy}{dt} = 0.05y$ with $y(0) = 40$. When does y reach 100?

Step 1. The question asks for a time rather than a value, so solve the particular solution for t instead of substituting into it.

Step 2. Separating gives $y = 40e^{0.05t}$; setting $40e^{0.05t} = 100$ gives $e^{0.05t} = 2.5$, so $t = 20 \ln 2.5$.
 $\Rightarrow \quad t \approx 18.326$

Try it: $\frac{dy}{dt} = 0.04y$ with $y(0) = 50$. When does y reach 150?

check: $t \approx 27.465$

(!) **Watch out:** A constant added *inside* an exponential is not the same as one added outside. From $\ln |y| = F(x) + C$ you get $y = e^{F(x)}e^C = Ae^{F(x)}$ — the constant multiplies, it does not add. Also: substitute the initial values *before* you rearrange, and never write a separate C on each side.